Ayman Mahfuz

Software Engineer | ML, Infrastructure, Systems & High-Performance Computing
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Education

The University of Texas at Austin — Austin, TX
Bachelor of Science in Computer Science — *Aug 2023 – May 2027*

- **Concentration:** AI and ML — **GPA:** 3.5/4.0
- **Courses:** Data Structures, Algorithms, High Performance Computing, Natural Language Processing, Generative Visual Computing, Energy-Efficient Computing

Experience

Arm — May 2025 – Aug 2026
Platform Validation Intern → Co-op – ML Infrastructure (Python, Bayesian Opt., PMU, Silicon)

- Reduced CPU and memory validation runs by more than **100x** while identifying stress configurations in the top **0.2%** of the search space by building a **Bayesian-optimization** system guided by performance monitoring unit (**PMU**) feedback
- Reproduced high-stress silicon failures across **three** CPU-platform validation phases by deploying the **ML** optimizer into Arm's production workflow and optimizing against cache-miss rate, memory bandwidth, and stall rate.
- Earned Arm patent-filing approval for the method and a 12-month co-op extension; presented production results to SVP and director leadership.

UT Austin - Texas Robotics & AI Lab — Jan 2025 – Jul 2026
Undergraduate Research Assistant – Reinforcement Learning (C++, Python, Multi-Agent RL, LLMs, GPU, CI)

- Helped UT Austin place **3rd** overall at RoboCup 2025 by deploying **15** hierarchical multi-agent **reinforcement learning** policies to physical **NAO robots** and building **C++/Python** training environments within a **400K+** line soccer stack
- Accelerated policy-training iterations by **2-3x** and scaled experiments to **5M+** episodes by profiling GPU utilization and eliminating training-pipeline bottlenecks.

UT Austin – Center for Media Engagement — Aug 2023 – May 2025
Software Engineer, Undergraduate Research Assistant (BigQuery, Selenium, REST, JavaScript, PyTorch, BERT)

- Recovered transient scraper and API failures while ingesting **50M+** articles and **70M+** comments into **250M+ BigQuery** rows by building concurrent **Selenium/REST** workers with retry handling.
- Prevented duplicate experiment events and preserved round-robin game assignments for **1,000+** participants through deterministic assignment logic, event deduplication, and structured logging.
- Achieved **99%** classification accuracy across clickbait, entity, sentiment, and story tasks by fine-tuning **BERT** models and iterating through validation and error analysis.

UT Austin – Oden Institute — Aug 2023 – Jan 2025
Undergraduate Research Assistant – Scientific ML Infrastructure (PyTorch, CNNs, Transformers, Apptainer, SLURM)

- Improved MRI-segmentation Dice score by **12** percentage points by benchmarking **convolutional** and **transformer** architectures across five-fold validation on **1,000+** scans using NVIDIA H100 GPUs
- Stabilized long-running **PyTorch** training and recovered failed jobs from checkpoints by resolving GPU-memory, I/O, and mixed-precision bottlenecks in an Apptainer/SLURM pipeline.

Projects

HelmPM (helmpm.app) – LLM-Native PM Platform | *React, FastAPI, PostgreSQL, OpenAI/Anthropic*

- Built multi-tenant React/FastAPI/PostgreSQL platform managing **thousands of tickets across 4 organizations**, with document/code retrieval and automated Linear/Jira sync.
- Engineered hybrid semantic/keyword retrieval and Merkle-tree indexing, enabling exact-ID search while re-embedding only changed documents.

253M-Parameter Transformer and RLHF Training Stack | *PyTorch, HPC*

- Built and pretrained **253M-parameter** PyTorch transformer on **600M tokens** using RoPE, RMSNorm, SwiGLU, and attention sinks.
- Engineered training stack with mixed precision, gradient accumulation, checkpoint recovery, evaluation, SFT, and DPO for reproducible multi-session HPC experiments.

LLM & Diffusion Energy Benchmark | *Python, PyTorch, vLLM, SLURM, GH200*

- Built Python/vLLM/SLURM energy benchmark using NVML and Zeus, running **577 tests across 55 configurations and 9 models** on GH200 GPUs.
- Measured **24–32% LLM energy reductions** from FP8/INT8 but zero diffusion savings; validated with Welch's t-tests/Bonferroni correction and explained via roofline analysis.

Skills

Languages: Python, C++, Java, TypeScript
Backend/Web/Data: React, FastAPI, Flask, PostgreSQL, BigQuery, REST APIs, Pandas, NumPy, AWS
ML/Systems: PyTorch, Transformers, vLLM, LoRA/PEFT, PPO/MAPPO, RLHF/DPO, SLURM, Docker, Linux, profiling, MCP, Harness Engineering, Zeus/NVML, HPC, Distributed Systems